# Reverse-check review package — `MSSMD_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

| item | value |
|---|---|
| model file | `MSSMD/repair/final.fr` |
| original model name | `MSSMD_gen` (hidden from the agent) |
| paper | MSSMD/text/1812.00380.txt |
| blindness scope | blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back |

## Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

**`LN1NDAD (:=)`**

```
I*gd*(neu1bar.Ga[mu].neuD)*AD[mu]
```

**`LADMuMu (:=)`**

```
gd*(mbar.Ga[mu].m)*AD[mu]
```

**`LTot (:=)`**

```
LN1NDAD + LADMuMu
```

## Blank-slate reconstruction

# Reconstructed Physics From `sanitized.fr`

## Lagrangian

No covariant derivatives `DC[...]` or field strengths `FS[...]` appear in this file. Therefore the file does not encode kinetic terms or gauge-covariant derivative structure. The only gauge/Lorentz content explicitly present is the vector field `AD[mu]` coupled to fermion vector currents through `Ga[mu] = \gamma^\mu`.

**`LN1NDAD`**

$$\mathcal{L}_{\texttt{LN1NDAD}} = i\, g_d\, \bar{\tilde{\chi}}_1^0 \gamma^\mu \tilde{\chi}_D\, A_{D\mu}$$

Here `neu1bar.Ga[mu].neuD` is the spinor contraction $\bar{\tilde{\chi}}_1^0 \gamma^\mu \tilde{\chi}_D$. Both `neu1` and `neuD` are declared self-conjugate Majorana fermions. No chiral projectors appear, so this is a pure vector-current Lorentz structure as written in the file. The explicit factor of $i$ is part of the `.fr` term.

**`LADMuMu`**

$$\mathcal{L}_{\texttt{LADMuMu}} = g_d\, \bar{\mu} \gamma^\mu \mu\, A_{D\mu}$$

Here `mbar.Ga[mu].m` is the muon vector current. The muon field `m` is used but not declared in this file, so it is inherited from whatever base model this file was intended to extend.

LTot

$$\mathcal{L}_{\texttt{LTot}} = \mathcal{L}_{\texttt{LN1NDAD}} + \mathcal{L}_{\texttt{LADMuMu}}$$

## Field Table

| .fr symbol | Members | Spin | SU(3) rep | SU(2) rep | U(1) / electric charge | Self-conjugate | Mass symbol and value |
|---|---|---|---|---|---|---|---|
| AD | AD | 1 | singlet, no color index | not declared | no Q declared, neutral by field usage | yes | MAD, no numeric value |
| neu | neu1, neu2, neu3, neu4 | 1/2 | singlet, no color index | not declared | no Q declared, neutral by names/PDG | yes | Mneu1 = 10., Mneu2, Mneu3, Mneu4 |
| ch | ch1, ch2 | 1/2 | singlet, no color index | not declared | Q = +1 for particle, antiparticle has Q = -1 | no | Mch1, Mch2, no numeric values |
| go | go | 1/2 | adjoint, Index[Gluon] | not declared | no Q declared | yes | Mgo, no numeric value |
| neuD | neuD | 1/2 | singlet, no color index | not declared | no Q declared, neutral by name/usage | yes | MneuD = 1. |
| h0 | h0 | 0 | singlet, no color index | not declared | no Q declared, neutral by name/PDG | yes | MH01 = 125. |
| H0 | H0 | 0 | singlet, no color index | not declared | no Q declared, neutral by name/PDG | yes | MH02, no numeric value |
| A0 | A0 | 0 | singlet, no color index | not declared | no Q declared, neutral by name/PDG | yes | MA0, no numeric value |
| Hp | H+, antiparticle H- | 0 | singlet, no color index | not declared | Q = +1 for particle, antiparticle has Q = -1 | no | MHp, no numeric value |
| sn | sn1, sn2, sn3 | 0 | singlet, no color index | not declared | Q = 0 | no | Msn1, Msn2, Msn3, no numeric values |

| .fr symbol | Members | Spin | SU(3) rep | SU(2) rep | U(1) / electric charge | Self-conjugate | Mass symbol and value |
|---|---|---|---|---|---|---|---|
| `sl` | `sl1-` … `sl6-` | 0 | singlet, no color index | not declared | `Q = -1` for particle, antiparticle has `Q = +1` | no | `Msl1` … `Msl6`, no numeric values |
| `su` | `su1` … `su6` | 0 | fundamental triplet, `Index[Colour]` | not declared | `Q = +2/3` for particle, antiparticle has `Q = -2/3` | no | `Msu1` … `Msu6`, no numeric values |
| `sd` | `sd1` … `sd6` | 0 | fundamental triplet, `Index[Colour]` | not declared | `Q = -1/3` for particle, antiparticle has `Q = +1/3` | no | `Msd1` … `Msd6`, no numeric values |

## Parameters

| Parameter | Type | Appears in | Meaning from file content |
|---|---|---|---|
| `gd` | external real scalar, value `0.001` | Multiplies `LN1NDAD` and `LADMuMu` | New vector-current coupling of `AD` to $\bar{\tilde{\chi}}_1^0\gamma^\mu\tilde{\chi}_D$ and $\bar{\mu}\gamma^\mu\mu$ |
| `RNN[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of a neutral-fermion mixing matrix, block `NMIX` |
| `INN[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of a neutral-fermion mixing matrix, block `IMNMIX` |
| `RUU[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of chargino $U$-type mixing matrix, block `UMIX` |
| `IUU[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of chargino $U$-type mixing matrix, block `IMUMIX` |
| `RVV[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of chargino $V$-type mixing matrix, block `VMIX` |
| `IVV[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of chargino $V$-type mixing matrix, block `IMVMIX` |
| `RRn[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of sneutrino mixing matrix, block `SNUMIX` |
| `IRn[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of sneutrino mixing matrix, block `IMSNUMIX` |

| Parameter | Type | Appears in | Meaning from file content |
|---|---|---|---|
| `RRl[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of charged-slepton mixing matrix, block `SELMIX` |
| `IRl[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of charged-slepton mixing matrix, block `IMSELMIX` |
| `RRu[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of up-squark mixing matrix, block `USQMIX` |
| `IRu[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of up-squark mixing matrix, block `IMUSQMIX` |
| `RRd[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Real part of down-squark mixing matrix, block `DSQMIX` |
| `IRd[i,j]` | external real matrix | Not used in any Lagrangian term in this file | Imaginary part of down-squark mixing matrix, block `IMDSQMIX` |
| `alp` | external real scalar | Not used in any Lagrangian term in this file | Scalar mixing angle or generic angle parameter, named $\alpha$, block `FRALPHA` |

### Physics Summary

The file declares an MSSM-like spectrum of neutralinos, charginos, gluino, Higgs-sector scalars, sleptons, sneutrinos, and squarks, plus an additional neutral vector boson `AD` and a neutral Majorana fermion `neuD`. The only interactions actually encoded are vector-current couplings of `AD` to the muon current and to a mixed Majorana current involving `neu1` and `neuD`. This mediates processes such as $\mu^+\mu^- \leftrightarrow A_D^* \leftrightarrow \tilde\chi_1^0\tilde\chi_D$, as well as `AD` production or decay through muons and the neutral fermion pair when kinematically allowed.

### Paper cross-check

## Comparison of `reconstruction.md` Against the Paper Model

### Located Paper Model Definitions

The paper does not provide a numbered Lagrangian or explicit symbolic field-representation table in the supplied `paper.tex`. The relevant model definitions are given in prose:

- **Section 1, Introduction:** dark SUSY benchmark: breaking of a new $U(1)_D$ gives a massive dark photon $\gamma_D$; $\gamma_D$ couples to SM particles through kinetic mixing $\epsilon$ with SM photons; topology $h \to 2n_1$, $n_1 \to n_D+\gamma_D$, $\gamma_D \to \mu^+\mu^-$.
- **Section 4, Signal modeling:** dark SUSY simulation choices: $h \to 2n_1$, $n_1 \to n_D + \gamma_D$, $m_{n_D} = 1$ GeV, $m_h = 125$ GeV, $m_{n_1} = 10$ GeV, $m_{\gamma_D} = 0.25$–$8.5$ GeV, $\gamma_D \to \mu^-\mu^+$ with 100% branching in the signal samples.
- **Section 7, Results / Figure 3 caption:** interpreted process $pp \to h \to 2n_1 \to 2\gamma_D + 2n_D \to 4\mu + X$, with $m_{n_1} = 10$ GeV and $m_{n_D} = 1$ GeV; limits shown in the $(\epsilon, m_{\gamma_D})$ plane.
- **NMSSM benchmark, Sections 1 and 4:** $h_{1,2} \to 2a_1$, followed by $a_1 \to \mu^+\mu^-$, with $h_{1,2}$ CP-even and $a_1$ CP-odd.

### Term-by-Term Comparison

| paper term (eq. ref) | reconstruction term | verdict | notes |
|---|---|---|---|
| Massive dark photon $\gamma_D$ from broken $U(1)_D$ (Section 1) | `AD`, spin-1, self-conjugate neutral vector, mass `MAD` | agree | The neutral massive vector field matches the paper's $\gamma_D$ at the level of particle content. The reconstruction does not encode the $U(1)_D$ gauge origin or symmetry breaking. |
| Kinetic mixing of $\gamma_D$ with SM photons via $\epsilon$ (Section 1; Section 7) | Direct coupling $g_d\,\bar{\mu}\gamma^\mu\mu\,A_{D\mu}$ | disagree | The paper defines the SM coupling mechanism as photon kinetic mixing controlled by $\epsilon$, while the reconstruction has a direct muon-only vector coupling with coefficient $g_d$. This may be an effective vertex, but it omits the photon mixing structure and expected coupling relation to electric charge. |
| Dark photon decay $\gamma_D \to \mu^-\mu^+$, set to 100% in signal samples (Section 4) | `LADMuMu`: $g_d\,\bar{\mu}\gamma^\mu\mu A_{D\mu}$ | agree | This term supports $\gamma_D \to \mu^+\mu^-$. The paper does not give the Lorentz structure explicitly, but a vector dark photon coupled to the electromagnetic current is consistent with a vector muon current. |
| Higgs cascade $h \to 2n_1$ (Sections 1 and 4; Figure 3 caption) | No $h\,n_1 n_1$ term | missing-in-reconstruction | The paper's dark SUSY topology requires Higgs decay to two lightest non-dark neutralinos. Reconstruction has scalar fields including `h0`, but no Higgs-neutralino interaction. |
| Neutralino transition $n_1 \to n_D + \gamma_D$ (Sections 1 and 4; Figure 3 caption) | `LN1NDAD`: $ig_d\,\bar{\tilde{\chi}}_1^0\gamma^\mu\tilde{\chi}_D A_{D\mu}$ | agree | This is the only reconstruction term matching the required $n_1 n_D \gamma_D$ transition. The paper does not specify chirality, phase convention, or coefficient structure, so the vector-current form cannot be fully validated from the paper text alone. |

| paper term (eq. ref) | reconstruction term | verdict | notes |
|---|---|---|---|
| Dark neutralino $n_D$, stable/undetected, $m_{n_D} = 1$ GeV (Section 4; Figure 3 caption) | `neuD`, self-conjugate neutral fermion, `MneuD = 1.` | agree | Mass and neutral invisible role match. The paper does not specify Majorana versus Dirac, so self-conjugacy is not confirmed by the paper text. |
| Lightest non-dark neutralino $n_1$, $m_{n_1} = 10$ GeV (Sections 1 and 4; Figure 3 caption) | `neu1`, self-conjugate neutral fermion, `Mneu1 = 10.` | agree | Mass and neutralino role match. As with $n_D$, the paper does not explicitly state Majorana conventions. |
| Dark photon mass range $m_{\gamma_D} = 0.25$–$8.5$ GeV (Section 4; Section 7) | `MAD`, no numeric value | missing-in-reconstruction | The reconstruction identifies a dark-vector mass parameter but does not encode the simulated mass range. |
| Dark photon lifetime controlled by $\epsilon$ and $m_{\gamma_D}$, $c\tau_{\gamma_D} = 0$–$100$ mm (Sections 1, 4, 7) | No lifetime or $\epsilon$-dependent width structure | missing-in-reconstruction | The implementation reconstruction lacks the kinetic-mixing parameter and the lifetime relation $\tau_{\gamma_D}(\epsilon, m_{\gamma_D}) = \epsilon^{-2} f(m_{\gamma_D})$. |
| NMSSM process $h_{1,2} \to 2a_1$, $a_1 \to \mu^+\mu^-$ (Sections 1 and 4) | No $h_{1,2}a_1a_1$ or $a_1\mu\mu$ terms; scalar table has `h0`, `H0`, `A0`, `Hp` | missing-in-reconstruction | The reconstruction does not encode the NMSSM benchmark interactions. `A0` could be a generic CP-odd scalar, but the paper's $a_1$ decay and Higgs cascade are absent. |
| CP-even Higgs bosons $h_1, h_2$ and CP-odd $a_1$ in NMSSM benchmark (Section 1) | `h0`, `H0`, `A0`, `Hp` scalar spectrum | disagree | The broad scalar spectrum resembles a two-Higgs-sector/MSSM-style list, but the paper specifically discusses NMSSM $h_1, h_2, a_1$, not charged Higgs phenomenology or the full scalar sector in the benchmark implementation. |
| Pair production of identical light bosons $pp \to 2a + X \to 4\mu + X$, model-independent framing (Section 1; Section 8) | `LTot = LN1NDAD + LADMuMu` only | missing-in-reconstruction | The reconstruction provides vertices for one dark SUSY decay chain but no production mechanism or generic pair-production operator. |

| paper term (eq. ref) | reconstruction term | verdict | notes |
|---|---|---|---|
| Full MSSM-like neutralinos `neu1`–`neu4`, charginos, gluino, sfermions, extra Higgs states | Reconstruction field table lists many MSSM-like fields | extra-in-reconstruction | The paper's dark SUSY interpretation only needs $h$, $n_1$, $n_D$, and $\gamma_D$, while the NMSSM benchmark needs $h_{1,2}$ and $a_1$. The extra MSSM-like spectrum is not defined in the paper's benchmark description. |
| Direct muon-only dark-vector coupling $g_d\,\bar{\mu}\gamma^{\mu}\mu A_{D\mu}$ | `LADMuMu` | extra-in-reconstruction | The paper states coupling to SM particles via kinetic mixing with photons, not a standalone muon-specific interaction. As an effective decay vertex it is useful, but as a model definition it is narrower than the paper. |
| Common coupling $g_d$ for $A_D\bar{\mu}\mu$ and $A_D\tilde{\chi}_1^0\tilde{\chi}_D$ | `gd` multiplies both `LN1NDAD` and `LADMuMu` | disagree | The paper distinguishes dark-sector dynamics from SM coupling through kinetic mixing $\epsilon$. It does not imply the same coefficient controls the neutralino transition and the muon coupling. |

## Disagreements and Human Checks

- **Kinetic mixing replaced by direct muon coupling — substantive**. Check whether `gd` was intended as an effective $\epsilon e Q_\mu$ coupling after diagonalizing kinetic mixing, or whether the implementation incorrectly omitted photon/dark-photon mixing.
- **Same `gd` used for dark-sector transition and muon coupling — substantive**. Check the original implementation or model card to see whether two physically distinct couplings were collapsed into one parameter.
- **Missing Higgs decay $h \to 2n_1$ — substantive**. Check whether production and Higgs decay were intentionally handled externally by BRIDGE/Pythia rather than encoded in the FeynRules model.
- **Missing $\epsilon$-dependent lifetime/width relation — substantive**. Check whether the dark photon lifetime was imposed at event-generation level rather than derived from the model file.
- **Missing NMSSM $h_{1,2} \to 2a_1$, $a_1 \to \mu^+\mu^-$ interactions — substantive**. Check whether the reconstruction is only for the dark SUSY benchmark and not meant to cover the NMSSM benchmark also discussed in the paper.
- **Scalar-sector mismatch: `h0`, `H0`, `A0`, `Hp` versus paper's $h_1, h_2, a_1$ — convention/substantive**. Check whether `A0` is intended to represent $a_1$, or whether this is a generic MSSM scaffold unrelated to the paper's NMSSM benchmark.
- **Large extra MSSM-like spectrum in reconstruction — convention**. Check whether these fields are inherited boilerplate from an MSSM/FeynRules template and unused in the actual generated process.
- **Majorana assignment for $n_1$ and $n_D$ — convention**. Check the underlying dark SUSY model reference or implementation, since the paper calls them neutralinos but does not explicitly state the Majorana convention in the supplied text.

## Overall Assessment

The reconstruction captures the core dark SUSY decay-chain fields $n_1$, $n_D$, and $\gamma_D$, including the benchmark masses $m_{n_1} = 10$ GeV and $m_{n_D} = 1$ GeV, and it includes vertices that can realize $n_1 \to n_D + \gamma_D$ and $\gamma_D \to \mu^+\mu^-$. However, it is not a faithful full representation of the model as described in the paper: the paper's defining SM portal is kinetic mixing through $\epsilon$, not a direct muon-only coupling; the Higgs production/decay step $h \to 2n_1$ is absent; the $\epsilon$-dependent lifetime structure is absent; and the NMSSM benchmark terms are not represented. The extra MSSM-like field content appears mostly unused and may be implementation scaffold rather than physics used in the paper's interpreted signal.

## Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 23 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

## Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: